

Achievable Acceleration Across the Fleet

Maximum linear and angular acceleration per body axis, by spacecraft class

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Contents

1	What these numbers are	1
2	Data coverage — and two kinds of zero	2
3	Per-vehicle acceleration	3
3.1	Boosters	3
3.2	Crewed Vehicles	3
3.3	GEO Satellites	3
3.4	LEO Satellites	3
3.5	Deep Space Probes	3
4	Typical values by class	5
5	What the classes actually look like	6
5.1	The rule-of-thumb table	6
6	Appendix A — signed per-axis detail	7
6.1	Boosters	7
6.2	Crewed Vehicles	7
6.3	GEO Satellites	7
6.4	LEO Satellites	7
6.5	Deep Space Probes	7
7	Limitations	9
8	Reproducing	9

1 What these numbers are

For each spacecraft the actuation-envelope model builds the **achievable force set** and the **achievable moment set** — every net wrench the vehicle’s thrusters and gimbaled engines can produce with all of them available at once. The maximum force along a body axis is the support function of that set evaluated on the axis, and the accelerations follow directly:

$$a_j = \frac{h_F(e_j)}{m} \quad \alpha_j = \frac{h_M(e_j)}{I_{jj}}$$

Each axis is evaluated in both directions and the **larger of the two** is reported here, since the question is what the vehicle can achieve. Layouts are often strongly asymmetric — a lander can push hard along $-z$ and barely at all along $+z$ — so the full signed breakdown is in Appendix A.

Three properties of this figure are worth being explicit about, because they make it an **upper bound** rather than an operating value:

- **Everything fires at once.** The number assumes every thruster and engine is simultaneously available and commanded to the optimum. No propellant budget, duty cycle, thermal limit or control-allocation loss is applied.
- **It is instantaneous.** Angular acceleration is M/I_{jj} about the principal axes, neglecting gyroscopic coupling and products of inertia, so it is a small-rate figure and not a slew capability.
- **Main engines are included.** For satellites the axial number is dominated by the apogee engine, which is a manoeuvre actuator and not something used for attitude or translation control. The lateral and angular figures are the ones that describe day-to-day control authority.

Masses and thrust levels are read from the fleet workbook; dimensions and inertias are shape-model estimates (uniform cylinder or box), which is the dominant uncertainty in every angular figure. A factor-of-two error in I_{jj} moves α_j by the same factor.

2 Data coverage — and two kinds of zero

A zero in these tables means one of two entirely different things, and pooling them would produce nonsense:

Structural zero. The modelled actuators genuinely produce no authority about that axis. A single centreline gimballed engine cannot generate roll no matter how large it is, which is why Ariane 5, Vega and Vega-C all show $\alpha_z = 0$ — real launchers solve this with a separate roll-control system that is not in the model. These are real results and are kept.

Data gap. The vehicle has the hardware, but the workbook records no thrust value for it. Reporting that as zero authority would be a statement about the spreadsheet, not the spacecraft.

Vehicle	Class	Why it is excluded
Sentinel-1A	LEO Satellites	no actuator thrust recorded
GOES-16 (GOES-R)	GEO Satellites	no thrust recorded for either tier
GOES-19 (GOES-U)	GEO Satellites	no thrust recorded for either tier

Those 3 carry no usable figure at all and are excluded from every statistic below.

The distinction is drawn **per quantity**, because a vehicle can be sound in one and empty in the other:

Vehicle	Linear usable	Angular usable	What is missing
Juno	yes	no	12 RCS thrusters named with no thrust recorded

Juno is the case that matters. Its main engine is documented, so its linear figure stands; but that engine is rigidly mounted on the centreline and produces no moment, which means its entire attitude authority is the twelve thrusters the workbook names and gives no thrust for. Its angular row is therefore not a measurement of Juno and is excluded from the angular statistics.

Note also what is **not** a gap: every booster here records its RCS tier as “None at vehicle level”, which is a design fact rather than a hole — a launcher of this kind controls attitude with engine gimbal alone. Europa Clipper likewise records “same engine set, no separate RCS tier”. Those vehicles are complete and are kept.

Usable vehicles per class:

Class	Linear	Angular
Boosters	4 of 4	4 of 4
Crewed Vehicles	4 of 4	4 of 4
GEO Satellites	2 of 4	2 of 4
LEO Satellites	3 of 4	3 of 4
Deep Space Probes	5 of 5	4 of 5

3 Per-vehicle acceleration

Linear acceleration in m/s^2 ; angular acceleration in $^\circ/\text{s}^2$. Both are the maximum over the two directions along each axis. Body axes follow each vehicle’s own convention as built in the model: z **is the main-engine / thrust axis**, and x, y are lateral. That is why the a_z column is almost always the largest — it is the main engine, and the lateral columns are what the RCS alone can do.

3.1 Boosters

Vehicle	Mass [kg]	a_x	a_y	a_z	α_x	α_y	α_z
Vega-C	210,000	2.51	2.51	22.14	24.4	24.4	0
Vega	137,000	2.43	2.43	21.47	27.6	27.6	0
Delta IV Heavy	725,700	1.35	1.35	12.91	6.6	6.6	34.0
Ariane 5 ECA	780,000	0.13	0.13	1.23	0.877	0.877	0

3.2 Crewed Vehicles

Vehicle	Mass [kg]	a_x	a_y	a_z	α_x	α_y	α_z
Crew Dragon (Dragon 2)	12,519	0.13	0.13	42.82	127.6	127.6	14.4
Apollo Lunar Module	15,103	0.41	0.41	3.21	153.1	153.1	34.4
Apollo Command & Service Module	30,320	0.28	0.28	3.07	7.2	7.2	3.4
Orion (CM + European Service Module)	26,536	0.14	0.14	1.06	4.1	4.1	2.2

3.3 GEO Satellites

Vehicle	Mass [kg]	a_x	a_y	a_z	α_x	α_y	α_z
Meteosat Second Generation (MSG)	2,000	0.0119	0.0107	0.41	0.640	9.7	1.2
Solar Dynamics Observatory (SDO)	3,100	0.0212	0.0154	0.18	0.463	0.463	2.2
GOES-16 (GOES-R) *	5,192	—	—	—	—	—	—
GOES-19 (GOES-U) *	5,000	—	—	—	—	—	—

3.4 LEO Satellites

Vehicle	Mass [kg]	a_x	a_y	a_z	α_x	α_y	α_z
SMAP	944	0.0141	0.0102	0.0133	0.960	0.960	2.1
Sentinel-3A	1,250	0.0024	0.0017	0.0022	0.237	0.115	0.210
GRACE-FO (per satellite)	600	3.3e-05	3.3e-05	1.7e-04	2.9e-03	5.6e-03	1.7e-03
Sentinel-1A *	2,300	—	—	—	—	—	—

3.5 Deep Space Probes

Vehicle	Mass [kg]	a_x	a_y	a_z	α_x	α_y	α_z
Juno *	3,625	0	0	0.18	—	—	—
Cassini (Cassini-Huygens)	5,574	0.0063	0.0063	0.16	0.240	1.2	0.256
New Horizons	478	0.0033	0.0033	0.0402	1.0	1.0	0.219
Europa Clipper	5,800	0	0	0.0379	0.539	0.520	0

Vehicle	Mass [kg]	a_x	a_y	a_z	α_x	α_y	α_z
Voyager 1	815	0.0041	0.0041	0.0058	0.949	0.949	1.0

4 Typical values by class

Median across the vehicles of each class with usable data, with the full range in brackets. The sample per class is small — this is a fleet survey, not a population — so the **range** is the honest statistic and the median is an indicative middle, not an expectation.

Class	n lin/ang	Axial a_z [m/s ²]	Lateral $a_{x,y}$ [m/s ²]	Pitch/yaw $\alpha_{x,y}$ [°/s ²]	Roll α_z [°/s ²]
Boosters	4/4	17.19 [1.23–22.14]	1.89 [0.13–2.51]	15.5 [0.877–27.6]	0 [0–34.0]
Crewed Vehicles	4/4	3.14 [1.06–42.82]	0.21 [0.13–0.41]	67.4 [4.1–153.1]	8.9 [2.2–34.4]
GEO Satellites	2/2	0.29 [0.18–0.41]	0.0165 [0.0119–0.0212]	5.1 [0.463–9.7]	1.7 [1.2–2.2]
LEO Satellites	3/3	0.0022 [1.7e-04–0.0133]	0.0024 [3.3e-05–0.0141]	0.237 [5.6e-03–0.960]	0.210 [1.7e-03–2.1]
Deep Space Probes	5/4	0.0402 [0.0058–0.18]	0.0033 [0–0.0063]	0.979 [0.539–1.2]	0.238 [0–1.0]

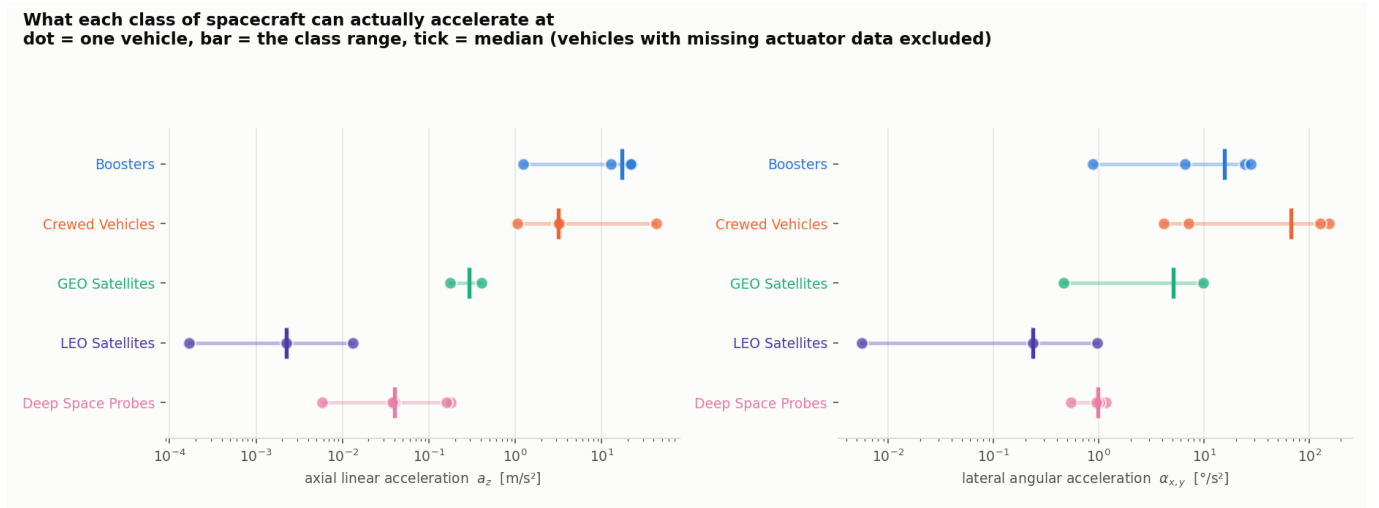


Figure 1: Acceleration by spacecraft class

5 What the classes actually look like

Axial acceleration separates the classes by four orders of magnitude, and it is really a statement about mission phase. Boosters sit around 17.19 m/s^2 because they must lift themselves against Earth gravity — anything below $\sim 10 \text{ m/s}^2$ does not leave the pad. Crewed vehicles come next at about 3.14 m/s^2 , sized for abort and orbital manoeuvring. GEO satellites (0.29 m/s^2) and deep-space probes (0.0402 m/s^2) carry an apogee or main engine sized for orbit insertion. LEO satellites are lowest of all (0.0022 m/s^2) because most carry no main engine at all: their axial figure *is* their RCS.

Lateral acceleration is about an order of magnitude below axial — and the exceptions say why. For every vehicle in the fleet that has a main engine, the axial-to-lateral ratio clusters tightly around **10:1** (interquartile range 8–12:1 over 13 vehicles). The reason is pure layout: the main engine points along one axis and everything sideways has to come from small RCS thrusters.

The vehicles that break the pattern break it in the direction that confirms it. Sentinel-3A, SMAP, Voyager 1 carry **no main engine at all**, and their ratios are 0.9:1, 0.9:1, 1.4:1 — near unity, because every axis including the nominal thrust axis is served by the same RCS. At the other extreme Crew Dragon reaches 340:1, its eight SuperDracos sized for launch abort against a Draco RCS sized for docking.

The practical consequence is the same either way: on any vehicle with a main engine, a guidance scheme that wants lateral authority is really asking the vehicle to tilt and use the main engine. That is a rotational manoeuvre with the associated delay, not a translation — which is exactly why the Apollo landing studies in this repository spend their control authority on attitude.

Angular acceleration does not track size or thrust; it tracks what the vehicle was built to do. The two largest figures in the fleet belong to the Apollo LM and Crew Dragon at 153 and $128 \text{ }^\circ/\text{s}^2$ — compact vehicles carrying RCS sized for landing and for docking. Boosters, with thrust in the meganewtons, manage $0.9\text{--}30 \text{ }^\circ/\text{s}^2$: their inertia is enormous and their only moment source is a few degrees of gimbal deflection on an engine close to the roll axis. Deep-space probes sit near $1.0 \text{ }^\circ/\text{s}^2$.

The rule that falls out: **a vehicle designed to manoeuvre precisely near something else has of order $100 \text{ }^\circ/\text{s}^2$; a vehicle designed to fly a trajectory has of order $1\text{--}30 \text{ }^\circ/\text{s}^2$.** Proximity operations are an angular-authority problem, and the fleet is sized accordingly.

Roll is the axis that gets sacrificed. Three of the four boosters have exactly zero roll authority from their modelled actuators, and Europa Clipper has none either. Roll is the axis a single centreline engine cannot touch, and in the fleet it is consistently the weakest or missing channel. Where a vehicle does have roll authority it usually comes from an offset engine cluster (Delta IV Heavy’s three boosters) or from a dedicated RCS ring.

5.1 The rule-of-thumb table

Rounded to the nearest order of magnitude, which is the precision these shape-model inertias actually support:

Class	Axial a_z	Lateral $a_{x,y}$	Pitch/yaw $\alpha_{x,y}$	Roll α_z
Boosters	$1\text{--}20 \text{ m/s}^2$	$0.1\text{--}3 \text{ m/s}^2$	$0.9\text{--}30 \text{ }^\circ/\text{s}^2$	$30 \text{ }^\circ/\text{s}^2$ — but 3 of 4 have none at all
Crewed Vehicles	$1\text{--}40 \text{ m/s}^2$	$0.1\text{--}0.4 \text{ m/s}^2$	$4\text{--}200 \text{ }^\circ/\text{s}^2$	$2\text{--}30 \text{ }^\circ/\text{s}^2$
GEO Satellites	$0.2\text{--}0.4 \text{ m/s}^2$	$0.01\text{--}0.02 \text{ m/s}^2$	$0.5\text{--}10 \text{ }^\circ/\text{s}^2$	$1\text{--}2 \text{ }^\circ/\text{s}^2$
LEO Satellites	$0.0002\text{--}0.01 \text{ m/s}^2$	$0.00003\text{--}0.01 \text{ m/s}^2$	$0.006\text{--}1.0 \text{ }^\circ/\text{s}^2$	$0.002\text{--}2 \text{ }^\circ/\text{s}^2$
Deep Space Probes	$0.006\text{--}0.2 \text{ m/s}^2$	$0.003\text{--}0.006 \text{ m/s}^2$	$0.5\text{--}1 \text{ }^\circ/\text{s}^2$	$0.2\text{--}1 \text{ }^\circ/\text{s}^2$ — but 1 of 4 have none at all

6 Appendix A — signed per-axis detail

The tables above take the larger of the two directions along each axis. Layouts are frequently asymmetric, and where they are, the *smaller* direction is the one that constrains a controller. Linear in m/s², angular in °/s².

6.1 Boosters

Vehicle	$+a_x$	$-a_x$	$+a_y$	$-a_y$	$+a_z$	$-a_z$	$+\alpha_x$	$-\alpha_x$	$+\alpha_y$	$-\alpha_y$	$+\alpha_z$	$-\alpha_z$
Vega-C	2.51	2.51	2.51	2.51	0	22.14	24.4	24.4	24.4	24.4	0	0
Vega	2.43	2.43	2.43	2.43	0	21.47	27.6	27.6	27.6	27.6	0	0
Delta IV Heavy	1.35	1.35	1.35	1.35	0	12.91	6.6	6.6	6.6	6.6	34.0	34.0
Ariane 5 ECA	0.13	0.13	0.13	0.13	0	1.23	0.877	0.877	0.877	0.877	0	0

6.2 Crewed Vehicles

Vehicle	$+a_x$	$-a_x$	$+a_y$	$-a_y$	$+a_z$	$-a_z$	$+\alpha_x$	$-\alpha_x$	$+\alpha_y$	$-\alpha_y$	$+\alpha_z$	$-\alpha_z$
Crew Dragon (Dragon 2)	0.13	0.13	0.13	0.13	0.18	42.82	127.6	127.6	127.6	127.6	14.4	6.0
Apollo Lunar Module	0.41	0.41	0.41	0.41	0.12	3.21	153.1	153.1	153.1	153.1	34.4	34.4
Apollo Command & Service Module	0.28	0.28	0.28	0.28	0.0587	3.07	7.2	7.2	7.2	7.2	3.4	3.4
Orion (CM + European Service Module)	0.14	0.14	0.14	0.14	0.0488	1.06	4.1	4.1	4.1	4.1	2.2	2.2

6.3 GEO Satellites

Vehicle	$+a_x$	$-a_x$	$+a_y$	$-a_y$	$+a_z$	$-a_z$	$+\alpha_x$	$-\alpha_x$	$+\alpha_y$	$-\alpha_y$	$+\alpha_z$	$-\alpha_z$
Meteosat Second Generation (MSG)	0.0029	0.0119	0.0041	0.0107	0.0098	0.41	0.569	0.640	9.7	9.5	1.2	0.293
Solar Dynamics Observatory (SDO)	0.0071	0.0212	0.0129	0.0154	0.0200	0.18	0.388	0.463	0.388	0.463	2.2	0.918

6.4 LEO Satellites

Vehicle	$+a_x$	$-a_x$	$+a_y$	$-a_y$	$+a_z$	$-a_z$	$+\alpha_x$	$-\alpha_x$	$+\alpha_y$	$-\alpha_y$	$+\alpha_z$	$-\alpha_z$
SMAP	0.0047	0.0141	0.0086	0.0102	0.0133	0.0133	0.804	0.960	0.804	0.960	2.1	0.895
Sentinel-3A	7.9e-04	0.0024	0.0014	0.0017	0.0022	0.0022	0.199	0.237	0.096	0.115	0.210	0.088
GRACE-FO (per satellite)	3.3e-05	3.3e-05	3.3e-05	3.3e-05	3.3e-05	1.7e-04	2.9e-03	2.9e-03	5.6e-03	5.6e-03	1.7e-03	1.7e-03

6.5 Deep Space Probes

Vehicle	$+a_x$	$-a_x$	$+a_y$	$-a_y$	$+a_z$	$-a_z$	$+\alpha_x$	$-\alpha_x$	$+\alpha_y$	$-\alpha_y$	$+\alpha_z$	$-\alpha_z$
Juno	0	0	0	0	0	0.18	—	—	—	—	—	—
Cassini (Cassini-Huygens)	0.0063	0.0063	0.0063	0.0063	1.0e-03	0.16	0.240	0.240	1.2	1.2	0.256	0.209
New Horizons	0.0033	0.0033	0.0033	0.0033	0.0033	0.0402	1.0	1.0	1.0	1.0	0.219	0.219

Vehicle	$+a_x$	$-a_x$	$+a_y$	$-a_y$	$+a_z$	$-a_z$	$+\alpha_x$	$-\alpha_x$	$+\alpha_y$	$-\alpha_y$	$+\alpha_z$	$-\alpha_z$
Europa Clipper	0	0	0	0	0	0.0379	0.539	0.539	0.520	0.520	0	0
Voyager 1	0.0041	0.0041	0.0041	0.0041	0.0058	0.0058	0.949	0.949	0.949	0.949	1.0	0.435

7 Limitations

- **Upper bound, not an operating value.** Every actuator fires at once, at full thrust, in the optimal direction. Real control allocation, duty cycles, propellant budgets and thermal limits all reduce these figures.
- **Inertia is a shape model.** Every vehicle is treated as a uniform cylinder or box at its published envelope dimensions. This is the largest single uncertainty in the angular columns and it is a *systematic* one: real spacecraft concentrate mass at the base, so true I_{xx}, I_{yy} are generally smaller than modelled and true angular accelerations correspondingly larger.
- **Actuator geometry is largely assumed.** Thrust magnitudes and counts come from the workbook; where thrusters sit and which way they point is a layout model (rings, face pairs, canted clusters), chosen per vehicle and recorded in the provenance sheet of `actuation_envelope_summary.xlsx`.
- **Small samples.** Two to five vehicles per class. The ranges are a survey of these particular spacecraft, not a distribution over the class.
- **Mass is a single reference value** — usually launch or wet mass. A vehicle near end of life with empty tanks accelerates considerably harder than these numbers suggest.
- **No gyroscopic coupling.** $\alpha_j = M_j/I_{jj}$ holds at low rates. For a spinning vehicle (Juno, or any spin-stabilised probe) the achievable angular acceleration about the transverse axes is materially different.

8 Reproducing

`python studies/actuation_envelopes/build_acceleration_tables.py`

Outputs `acceleration_reference.pdf`, `acceleration_reference.csv` and `figures/_acceleration_by_category.png` in this directory. The underlying envelope model, its per-value provenance flags and the three-panel per-vehicle figures are produced by `build_report.py` alongside it.